

CPTG Native-Phase Structural Test

Effective-Radius Closure and Structural Mode N of ZF-COS-20115

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Abstract

ZF-COS-20115 (ZF-COSMOS-20115) is a massive compact quiescent galaxy observed at $z = 3.715$. The adopted source set gives stellar velocity dispersion $\sigma_\star = 283 \pm 52 \text{ km s}^{-1}$, semimajor-axis effective radius $R_e = 0.66 \pm 0.08 \text{ kpc}$, stellar mass $1.15_{-0.09}^{+0.16} \times 10^{11} M_\odot$, and half-mass formation marker $z_{\text{form}} = 6.1_{-0.7}^{+0.8}$; the measured Sérsic structure and flattening provide the radial source geometry. Redshift selects CPTG native phase $N_{\pi, \text{obs}} = -1.55075$, while the formation marker maps to $N_{\pi, \text{form}} = -1.96009$. Native phase accumulates 4.57 times faster per gigayear from formation to observation than from observation to the normalized present coordinate. Re-expressing the source distances on the symmetric CPTG acoustic/luminosity midpoint gives $R_e = 0.65350 \text{ kpc}$, and the declared one-radius pressure-support reduction yields $a_\star \simeq 3.9717 \times 10^{-9} \text{ m s}^{-2}$. The stellar source gives $g_{\text{CPTG}, e} = 2.1791 \times 10^{-8} \text{ m s}^{-2}$, within 6.4% of the correspondingly converted published virial comparator. Structural Mode gives $N_{\text{sph}} = 3.40287$ and $N_{\text{obl}} = 3.45081$. The results provide a native-phase timing measurement, shared-input effective-radius closure, and radial structural classification without a dark-halo term or a redshift-driven $a_\star(t)$ law, while placing the galaxy's rapid early assembly and quenching in direct contrast with the halo-plus-feedback histories required by standard Λ CDM galaxy-formation models.

1 Introduction: Why ZF-COS-20115 Matters

ZF-COS-20115 is a stringent high-redshift galaxy because several normally separate formation requirements are already satisfied in the same compact system. It is observed at $z = 3.715$, contains of order $10^{11} M_\odot$ in stars, has a sub-kiloparsec effective radius, and is quiescent. Spectroscopy and imaging establish a large stellar velocity dispersion, a concentrated Sérsic profile, and a dynamical mass comparable to the independently inferred stellar mass [1, 2, 3].

The adopted stellar-population reconstruction also places half of the formed stellar mass by $z_{\text{form}} = 6.1_{-0.7}^{+0.8}$ [1]. The object therefore supplies both a native-history timing interval and a compact bound-system state that can be evaluated at the selected observed phase. Its rapid early assembly and quiescence also make it a well-studied stress case for dark-matter-based galaxy formation, a comparison developed below [2, 4, 5].

The calculation keeps cosmological placement and galaxy structure separate: redshift selects the native π -branch phase and its comparison coordinates, after which the measured galaxy state determines the object-dependent a_* , effective-radius field closure, and Structural Mode N .

2 Purpose, Test Definition, and Observed Inputs

The operational sequence is [7, 8]

$$\begin{array}{c}
 \underbrace{z \longrightarrow N_\pi \longrightarrow \text{selected epoch and branch coordinates}}_{\text{cosmological placement}} \\
 \Downarrow \\
 \underbrace{\text{galaxy structure} \longrightarrow a_* \longrightarrow \{\text{field closure, } N, \text{ CSMI}\}}_{\text{structural measurement}}
 \end{array} \tag{1}$$

The first line answers when the galaxy is represented in the CPTG geometry. The second asks what structural state the galaxy itself measures at that selected phase. No factor of $H(z)$ or $E(z)$ is applied to evolve the intrinsic a_* , and no present-equivalent $a_*(t)$ trajectory is introduced.

2.1 Observed inputs and source distance convention

The adopted source set is the internally matched structural and kinematic measurement set reported for ZF-COS-20115 by Esdaile et al. [1]. Their structural analysis fits single-component Sérsic profiles to HST/WFC3 F160W imaging and reports the semimajor-axis half-light radius, Sérsic index, and projected axis ratio $q = b/a$. The same analysis provides the aperture-corrected stellar velocity dispersion and the virial dynamical mass. The deblended stellar-population measurements supply the stellar mass and formation marker [1, 3].

The source convention is flat Λ CDM with

$$H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_m = 0.3, \quad \Omega_\Lambda = 0.7. \tag{2}$$

These quantities identify the distance convention used to report physical sizes and masses; they do not govern the CPTG branch mathematics.

The present acoustic and luminosity CPTG comparison endpoints are

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351, \quad H_{0,\text{lum}}^{\text{CPTG}} = 73.17 \text{ km s}^{-1} \text{ Mpc}^{-1}. \tag{3}$$

For a neutral source-unit conversion, this paper uses the arithmetic midpoint of the acoustic and luminosity *distance coordinates*, not a new cosmological branch. For $b \in \{\text{ac, lum}\}$,

$$D_{M,b}(z) = \frac{c}{H_{0,b}} \int_0^z \frac{dz'}{E_b(z')}, \quad D_{A,b}(z) = \frac{D_{M,b}(z)}{1+z}, \tag{4}$$

and

$$D_{A,\text{mid}}(z) \equiv \frac{D_{A,\text{ac}}(z) + D_{A,\text{lum}}(z)}{2}. \quad (5)$$

At $z = 3.715$,

$$D_A^{\Lambda 70} = 1477.0293 \text{ Mpc}, \quad (6)$$

$$D_{A,\text{ac}} = 1519.2743 \text{ Mpc}, \quad (7)$$

$$D_{A,\text{lum}} = 1405.7115 \text{ Mpc}, \quad (8)$$

$$D_{A,\text{mid}} = 1462.4929 \text{ Mpc}, \quad (9)$$

so

$$s_D \equiv \frac{D_{A,\text{mid}}}{D_A^{\Lambda 70}} = 0.99015833. \quad (10)$$

The corresponding source-coordinate transformations are

$$R_e = 0.660 \pm 0.080 \text{ kpc} \longrightarrow 0.65350 \pm 0.07921 \text{ kpc}, \quad (11)$$

$$M_\star = 1.15^{+0.16}_{-0.09} \times 10^{11} M_\odot \longrightarrow 1.1275^{+0.1569}_{-0.0882} \times 10^{11} M_\odot, \quad (12)$$

$$M_{\text{dyn}} = (0.72 \pm 0.27) \times 10^{11} M_\odot \longrightarrow (0.7129 \pm 0.2673) \times 10^{11} M_\odot. \quad (13)$$

The stellar mass uses the leading fixed-redshift scaling $M_\star \propto D_L^2$, while the source virial mass scales linearly with R_e at fixed $\sigma_\star$ and Sérsic coefficient.

Quantity	Adopted CPTG-coordinate value	Role in test
Formal target	ZF-COS-20115	Object identity
Redshift	$z = 3.715$	Common cosmological phase
Formation marker	$z_{\text{form}} = 6.1^{+0.8}_{-0.7}$	Half-mass native-phase marker
Stellar velocity dispersion	$283 \pm 52 \text{ km s}^{-1}$	Effective-radius support measurement
F160W effective radius	$0.65350 \pm 0.07921 \text{ kpc}$	Semimajor-axis half-light scale
Sérsic index	$n = 4.10 \pm 0.99$	Radial source shape
Projected axis ratio	$q_{\text{proj}} = 0.61 \pm 0.07$	Oblate-source geometry constraint
Total stellar mass	$1.1275^{+0.1569}_{-0.0882} \times 10^{11} M_\odot$	Radial stellar-source normalization
Published virial dynamical mass	$(0.7129 \pm 0.2673) \times 10^{11} M_\odot$	Shared-input closure comparator

Table 1: Inputs used in the ZF-COS-20115 calculation. Distance-dependent source quantities are re-expressed on the symmetric CPTG acoustic/luminosity midpoint distance coordinate before the bound-system reduction.

3 Native Phase and Branch Coordinates

Redshift selects one common native π -branch phase. The acoustic and luminosity quantities below are projections of that same native history, so their different integrated clock coordinates are readouts of one geometric phase rather than different galaxies or independent cosmologies.

For the observed redshift,

$$a = \frac{1}{1+z} = 0.2120890774, \quad N_\pi = \ln a = -1.550748916. \quad (14)$$

The native π branch and its acoustic and luminosity projections are [7, 8]

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \quad \mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}, \quad (15)$$

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad p_{\text{ac}} = 3 - \frac{\pi}{100}, \quad (16)$$

$$E_{\text{lum}}(a)^2 = \Omega_{\Lambda, \text{ac}} + \Omega_{m, \text{ac}} a^{-p_{\text{ac}}}, \quad \Omega_{m, \text{ac}} = \frac{p_{\text{ac}}}{3\pi}, \quad \Omega_{\Lambda, \text{ac}} = 1 - \Omega_{m, \text{ac}}. \quad (17)$$

The present normalizations are

$$H_0^{(\pi)} = 69.4162507897, \quad H_{0, \text{ac}}^{\text{CPTG}} = 67.4777967351, \quad H_{0, \text{lum}}^{\text{CPTG}} = 73.17 \quad (18)$$

in $\text{km s}^{-1} \text{Mpc}^{-1}$. At the observed phase,

$$E_\pi = 6.499212, \quad E_{\text{ac}} = 5.697409, \quad E_{\text{lum}} = 5.668422. \quad (19)$$

For any branch of the form $E^2 = B + Ta^{-q}$, the integrated branch-clock coordinate is

$$t_b(a) = \frac{2}{qH_{0,b}\sqrt{B}} \sinh^{-1} \left(\sqrt{\frac{B}{T}} a^{q/2} \right). \quad (20)$$

This gives

Clock coordinate	$t(z = 3.715)$ (Gyr)	t_0 (Gyr)
Native π branch	1.387259	12.754121
Acoustic projection	1.725668	13.885174
Luminosity comparison	1.599732	12.843057

Table 2: Branch-specific clock coordinates at the common ZF-COS-20115 geometric phase and at the normalized present phase.

4 Early Galaxy Challenge and CPTG Resolution

The phrase “impossible early galaxy” is useful only if the underlying problem is stated precisely. ZF-COS-20115 is not mathematically forbidden by Λ CDM, and a single galaxy does not by itself falsify cold dark matter. The difficulty is the complete state already reached by the observed epoch. By $z = 3.715$, a compact $\sim 10^{11} M_\odot$ stellar system has assembled and is already quiescent; the adopted stellar-population reconstruction places its half-mass formation marker at

$$z_{\text{form}} = 6.1^{+0.8}_{-0.7}, \quad (21)$$

while later gas observations give a conservative molecular-gas fraction limit $f_{\text{mol}} < 3.2\%$ [2, 1, 6]. The challenge is therefore the compressed sequence of halo and baryonic assembly,

efficient stellar-mass growth, compactification, and suppression of further star formation on a short early-Universe timetable.

In the standard Λ CDM galaxy-formation picture, that history is built inside a cold-dark-matter halo. Early studies found no matching object in several then-current hydrodynamical calculations, while later semi-analytic work demonstrated that rare analogues can be produced [2, 4, 5]. Those analogues require unusually early massive-halo growth, substantial merger-driven assembly, rapid central black-hole growth, and feedback strong enough to suppress subsequent gas cooling and star formation. The Meraxes analogues reach halo masses of order $10^{13} M_{\odot}$ and use rapid black-hole growth and AGN heating to quench the system; the same calculations show that halo angular momentum can materially alter the modeled gas-collapse history [5]. The issue is therefore the efficiency and timing of the coupled halo–baryon–feedback sequence, not the literal impossibility of a massive halo at this redshift.

CPTG treats the same observations differently. Redshift selects the common native phase and the source-inferred formation marker identifies an earlier point on that history. For $z_{\text{form}} = 6.1$,

$$N_{\pi,\text{form}} = -1.960094784, \quad t_{\pi,\text{form}} = 0.730734 \text{ Gyr}, \quad (22)$$

while the observed phase is $N_{\pi,\text{obs}} = -1.550748916$ at $t_{\pi,\text{obs}} = 1.387259 \text{ Gyr}$. The average native-phase advance is $0.623504 \text{ Gyr}^{-1}$ from formation to observation and $0.136427 \text{ Gyr}^{-1}$ from observation to the normalized present coordinate, giving

$$\mathcal{R}_{\text{cadence}} = \frac{\langle dN_{\pi}/dt_{\pi} \rangle_{\text{form} \rightarrow \text{obs}}}{\langle dN_{\pi}/dt_{\pi} \rangle_{\text{obs} \rightarrow 0}} = 4.57023, \quad (23)$$

with the quoted formation-marker range giving approximately 4.25–4.92. The gravitational accounting also differs: CPTG inserts no dark-halo term into the galaxy-scale field solve; once phase is selected, the measured baryonic source and object-dependent $a_{\star}$ determine the bound-system response tested below. Thus the CPTG resolution claimed here is a higher early native cadence together with recovery of the measured gravitational state from a baryonic structural field solve. CPTG does not thereby replace the microphysics of star formation, gas cooling, chemical enrichment, black-hole growth, or quenching; those remain astrophysical processes to be constrained observationally.

5 Effective-Radius Structural Closure

The cosmological placement does not determine $a_{\star}$. At the selected observed phase, the galaxy supplies the structural measurement itself.

5.1 Pressure-support measurement

The measured stellar dispersion and converted effective radius define

$$a_{\sigma}(R_e) = \frac{\sigma_{\star}^2}{R_e}. \quad (24)$$

The CPTG core framework does not identify a_σ and $a_\star$ by universal definition. In this declared one-radius pressure-supported reduction, $a_\sigma(R_e)$ is adopted as the estimator of the object-dependent structural acceleration scale. Thus

$$a_\star \simeq a_\sigma = \frac{(283 \text{ km s}^{-1})^2}{0.65350 \text{ kpc}} = (3.9717 \pm 1.5369) \times 10^{-9} \text{ m s}^{-2}, \quad (25)$$

where the displayed uncertainty propagates the quoted $\sigma_\star$ and R_e errors in quadrature.

The corresponding lowercase bound-system curvature-transport scale is

$$R_c = \frac{48\pi c^2}{a_\star} = 110.59 \text{ Gpc}. \quad (26)$$

Uppercase R_C is reserved for the universe-scale native cosmological curvature-transport radius and is not used for this galaxy reduction.

5.2 CPTG field solve and virial comparator

The source paper does not tabulate a separate aperture stellar mass. Under the same spatially constant stellar mass-to-light assumption used for the radial Sérsic reconstruction below, the semimajor-axis half-light radius contains one half of the model stellar mass. The one-radius stellar source is therefore

$$M_{\star,e} = \frac{1}{2} M_\star = 5.6374 \times 10^{10} M_\odot, \quad (27)$$

which gives

$$g_{\star,e} = \frac{GM_{\star,e}}{R_e^2} = 1.83994 \times 10^{-8} \text{ m s}^{-2}. \quad (28)$$

This is an aperture-source construction under the declared constant- M/L reduction, not a separately fitted aperture mass.

The reduced CPTG response is solved from

$$g = g_\star + \frac{\sqrt{a_\star g_\star} + g_{\text{geom}}}{\left[1 + (g/a_\star)^{123/50}\right]^{11/50}}, \quad (29)$$

with

$$g_{\text{geom}}(r) = \frac{1}{2} a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}. \quad (30)$$

At $r = R_e$,

$$g_{\text{CPTG,e}} = 2.17905 \times 10^{-8} \text{ m s}^{-2}. \quad (31)$$

Applying the quoted stellar-mass, dispersion, and radius perturbations one at a time through the full implicit solve and combining their half-ranges in quadrature gives a

descriptive local sensitivity of approximately

$$\delta g_{\text{CPTG,e}} = 0.5547 \times 10^{-8} \text{ m s}^{-2}. \quad (32)$$

At this compact radius the explicit finite- R_c term is only

$$g_{\text{geom}}(R_e) = 6.00 \times 10^{-21} \text{ m s}^{-2}, \quad (33)$$

so the one-radius closure is controlled by the stellar source and the $\sqrt{a_\star g_\star}$ curvature-polarization/transport response rather than by the finite- R_c contribution itself.

The correspondingly converted published virial comparator gives

$$g_{\text{dyn,e}} = \frac{GM_{\text{dyn}}}{R_e^2} = 2.32682 \times 10^{-8} \text{ m s}^{-2}. \quad (34)$$

Hence

$$g_{\text{CPTG,e}} - g_{\text{dyn,e}} = -1.478 \times 10^{-9} \text{ m s}^{-2}, \quad (35)$$

a central field-amplitude difference of about -6.4% .

In mass space the CPTG field corresponds to

$$\log_{10}(M_{\text{CPTG,e}}/M_\odot) = 10.82454, \quad (36)$$

compared with

$$\log_{10}(M_{\text{dyn,e}}/M_\odot) = 10.85304, \quad (37)$$

so

$$\Delta \log_{10} M_e = -0.02850 \text{ dex}. \quad (38)$$

This is a shared-input closure result, not an independent prediction significance. The source virial mass itself is constructed from $\sigma_\star$, R_e , and the Sérsic-dependent coefficient

$$\beta(n) = 8.87 - 0.831n + 0.0241n^2, \quad (39)$$

for which $\beta(4.10) = 5.8680$ [1]. The comparison therefore asks whether the CPTG field constructed from the same observed galaxy state remains consistent with the published pressure-supported dynamical scale.

6 Structural Mode N

The effective-radius closure tests field amplitude at one radius. Structural Mode asks where the solved radial field is organized. The construction follows the dedicated CPTG Structural Mode study [9]:

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad \kappa_{\text{struct}} = 1 \text{ kpc}, \quad (40)$$

$$w(r) = \left| \frac{dC_g}{dr} \right|, \quad (41)$$

and

$$\lambda = \frac{\int_{R/5}^R r w(r) dr}{\int_{R/5}^R w(r) dr}, \quad N = \frac{R}{\lambda}. \quad (42)$$

The cosmological phase N_π and Structural Mode N are distinct: N_π specifies when the galaxy is represented, while N specifies where its solved field is radially organized.

Because an ideal Sérsic profile has no finite outer edge, the registered outer readout radius is the projected semimajor-axis radius enclosing 95 percent of the model light. For Sérsic index n ,

$$\frac{L(< R)}{L_{\text{tot}}} = \frac{\gamma(2n, b_n(R/R_e)^{1/n})}{\Gamma(2n)}, \quad \frac{\gamma(2n, b_n)}{\Gamma(2n)} = \frac{1}{2}, \quad (43)$$

and R_{95} is defined by setting the enclosed fraction to 0.95. With $n = 4.10$ and $R_e = 0.65350 \text{ kpc}$,

$$R = R_{95} = 5.80731 \text{ kpc}. \quad (44)$$

6.1 Spherical-equivalent baseline

The baseline Abel-deprojects the measured Sérsic profile as a spherical-equivalent constant- M/L stellar source,

$$g_{\text{bar,sph}}(r) = \frac{GM_\star(< r)}{r^2}, \quad (45)$$

and solves the same CPTG response law over radius using the observed-epoch $a_\star$ in Eq. (25). The central result is

$$\lambda_{\text{sph}} = 1.70659 \text{ kpc}, \quad N_{\text{sph}} = 3.40287. \quad (46)$$

Under the adopted CSMI intervals, $3.30 < N \leq 3.55$ maps to Bulged Spiral. The continuous mode lies 0.10287 above the $N = 3.30$ boundary.

The quoted source uncertainties give the one-at-a-time sensitivity shown in Table 3.

Structural case	N_{sph}	CSMI readout
Central source	3.40287	Bulged Spiral
$\sigma_{\star} = 231 \text{ km s}^{-1}$	3.48698	Bulged Spiral
$\sigma_{\star} = 335 \text{ km s}^{-1}$	3.33721	Bulged Spiral
$n = 3.11$	3.45485	Bulged Spiral
$n = 5.09$	3.30866	Bulged Spiral
$M_{\star} = 1.0392 \times 10^{11} M_{\odot}$	3.38633	Bulged Spiral
$M_{\star} = 1.2843 \times 10^{11} M_{\odot}$	3.42980	Bulged Spiral
$R_e = 0.57429 \text{ kpc}$	3.47877	Bulged Spiral
$R_e = 0.73272 \text{ kpc}$	3.33785	Bulged Spiral

Table 3: Observed-epoch Structural Mode sensitivity under the spherical-equivalent Sérsic reconstruction. Each row varies one quoted source quantity at a time while retaining the same declared reconstruction.

Grid convergence from 4000 to 32000 radial support points gives $N = 3.402871047$, 3.402870923 , 3.402870946 , and 3.402870947 , respectively. The central mode and its placement above $N = 3.30$ are therefore not radial-grid artifacts.

6.2 Measured flattening: oblate-source geometry stress test

The spherical baseline does not use the measured projected axis ratio. The source fit gives

$$q_{\text{proj}} = \frac{b}{a} = 0.61 \pm 0.07. \quad (47)$$

For a family of similar oblate stellar-density surfaces

$$m^2 = R^2 + \frac{z^2}{q_{\text{int}}^2}, \quad 0 < q_{\text{int}} \leq 1, \quad (48)$$

observed at inclination i [10],

$$q_{\text{proj}}^2 = \cos^2 i + q_{\text{int}}^2 \sin^2 i. \quad (49)$$

Thus the central measured $q_{\text{proj}} = 0.61$ admits

$$52.4105^\circ \lesssim i \leq 90^\circ. \quad (50)$$

The edge-on endpoint has $q_{\text{int}} = q_{\text{proj}} = 0.61$, while $q_{\text{int}} \rightarrow 0$ as the inclination approaches the lower geometric limit.

Let $S(s)$ denote the measured semimajor-axis Sérsic shape. With constant stellar M/L , the corresponding similar-oblate density can be normalized to the same total stellar mass, and the exact equatorial source field of nested oblate homoeoids is [10]

$$g_{\text{bar,obl}}(R) = \frac{4\pi G q_{\text{int}}}{R} \int_0^R \frac{\rho_q(m) m^2 dm}{\sqrt{R^2 - (1 - q_{\text{int}}^2) m^2}}. \quad (51)$$

The CPTG response law itself is unchanged; only the baryonic source field is replaced. This is an axisymmetric source-geometry stress test, not a Jeans solution and not an inferred rotation curve.

At the edge-on central endpoint,

$$\lambda_{\text{obl}} = 1.68288 \text{ kpc}, \quad N_{\text{obl}} = 3.45081. \quad (52)$$

The measured flattening therefore shifts the structural centroid inward by about 0.0237 kpc and raises the mode by

$$\Delta N = N_{\text{obl}} - N_{\text{sph}} = 0.04794. \quad (53)$$

The central inclination/thickness family is summarized in Table 4.

Inclination	q_{int}	N	CSMI readout
90°	0.61000	3.45081	Bulged Spiral
80°	0.59378	3.45301	Bulged Spiral
70°	0.53751	3.46081	Bulged Spiral
60°	0.40348	3.48041	Bulged Spiral
55°	0.25347	3.50427	Bulged Spiral
53°	0.12470	3.52667	Bulged Spiral
Razor-thin numerical limit	$\rightarrow 0$	3.55040	Boundary-level

Table 4: Non-fitted oblate-source geometry family for the central measured source. The $q_{\text{int}} \rightarrow 0$ limit lies essentially on the adopted $N = 3.55$ Bulged-Spiral/Lenticular-Early-Disk boundary, so continuous N remains the primary structural result.

At the edge-on endpoint, one-at-a-time source variations give $N = 3.38328$ – 3.53411 across the quoted stellar-dispersion interval, 3.34529 – 3.51127 across the quoted Sérsic-index interval, 3.43403 – 3.47783 across the stellar-mass interval, 3.38528 – 3.52610 across the effective-radius interval, and 3.44151 – 3.46046 across the measured axis-ratio interval. Across these variations the measured flattening raises N relative to the spherical-equivalent source; the downstream CSMI label becomes boundary-sensitive only near the very thin limit.

At fixed source quadrature, changing the radial support grid from 4000 to 32000 points changes the central edge-on result by less than 2×10^{-10} in N . Numerical discretization is negligible compared with the observational and light-traces-mass systematics.

7 Results, Interpretation, and Limitations

The calculation establishes three internal CPTG results for the same galaxy and one external comparison result. The external comparison is that ZF-COS-20115 remains a demanding Λ CDM dark-matter galaxy-formation case: successful analogues require an unusually early combination of massive-halo assembly, mergers, rapid black-hole growth, and strong quenching feedback, even though such analogues are not forbidden. First within CPTG, the observed redshift places ZF-COS-20115 at $N_{\pi, \text{obs}} = -1.550748916$ on

the common native phase, while the source half-mass formation marker gives $N_{\pi,\text{form}} = -1.960094784$. The average native phase cadence from formation to observation is 4.57 times the observation-to-present cadence, with the quoted formation-marker range giving approximately 4.25–4.92.

Second, the selected observed galaxy state gives $a_\star \simeq 3.9717 \times 10^{-9} \text{ m s}^{-2}$ in the declared one-radius pressure-supported reduction. Solving the CPTG field with the constant- M/L half-mass stellar source gives $g_{\text{CPTG,e}} = 2.1791 \times 10^{-8} \text{ m s}^{-2}$, compared with the correspondingly converted published virial field $g_{\text{dyn,e}} = 2.3268 \times 10^{-8} \text{ m s}^{-2}$. The -6.4% field difference and -0.0285 dex mass difference are shared-input closure measures, not independent prediction significances.

Third, the radial reconstruction gives $N_{\text{sph}} = 3.40287$ for the spherical-equivalent source and $N_{\text{obl}} = 3.45081$ for the central edge-on oblate source. All listed one-at-a-time spherical source variations remain above $N = 3.30$. The measured flattening moves N upward; the central inclination family approaches the $N = 3.55$ CSMI boundary only in the razor-thin limit. Continuous N , not the discrete catalog label, therefore carries the primary structural result.

The main physical systematic is the stellar mass distribution. The source structural fit uses F160W light, corresponding to rest-frame near-UV at this redshift, and the source paper explicitly notes that this light need not perfectly trace stellar mass if radial M/L gradients are present [1]. The present Structural Mode calculation is therefore conditional on a spatially constant stellar M/L . A resolved rest-frame optical or near-infrared stellar-mass profile can replace that source assumption directly without changing the definitions of λ or N .

Resolved stellar kinematics also remain unavailable. The integrated $\sigma_\star$ does not uniquely separate random pressure support from ordered rotation, so the oblate calculation is intentionally a source-geometry stress test rather than a full axisymmetric dynamical solution. Spatially resolved stellar velocities would permit a stronger support reconstruction and would test whether a rotation-aware estimator is required for this object.

Finally, the branch clocks have only a cosmological placement and native-history role in the bound-system calculation. They do not define an $a_\star(t)$ law and do not rescale the measured object-dependent $a_\star$ merely because ZF-COS-20115 is observed at high redshift.

8 Conclusion

ZF-COS-20115 provides a compact high-redshift test in which cosmological placement, effective-radius field support, and radial structural organization are evaluated separately. The observed and half-mass formation markers give $N_{\pi,\text{obs}} = -1.550748916$ and $N_{\pi,\text{form}} = -1.960094784$, with a formation-to-observation native cadence 4.57 times the observation-to-present cadence. At the observed phase, the one-radius reduction gives $a_\star \simeq (3.9717 \pm 1.5369) \times 10^{-9} \text{ m s}^{-2}$; the CPTG field $g_{\text{CPTG,e}} = 2.1791 \times 10^{-8} \text{ m s}^{-2}$ differs from the converted published virial comparator $g_{\text{dyn,e}} = 2.3268 \times 10^{-8} \text{ m s}^{-2}$ by -0.0285 dex in equivalent mass, a shared-input closure rather than an independent significance test. The radial reconstruction gives $N_{\text{sph}} = 3.40287$ and $N_{\text{obl,edge}} = 3.45081$, with the measured

flattening strengthening rather than weakening the high Structural Mode.

The same galaxy remains an external stress case for standard dark-matter-based galaxy formation: published Λ CDM analogues require rare early halo histories involving rapid assembly, significant mergers, accelerated central black-hole growth, and sufficiently strong AGN feedback to suppress continued cooling and star formation [4, 5]. CPTG does not require that dark-halo gravitational contribution in the galaxy field solve; within the declared constant- M/L , one-radius pressure-support reduction, the galaxy is consistently placed on the native phase, closes against the published effective-radius virial scale, and remains a high- N structural system when its observed flattening is inserted explicitly into the source geometry. Formation and quenching microphysics remain separate astrophysical questions.

Data, Code, and Evidence Availability

The observational inputs used in this study are drawn from the cited literature, principally Esdaile et al. for the internally matched stellar velocity dispersion, F160W structural parameters, virial dynamical mass, stellar mass, and half-mass formation marker of ZF-COS-20115. The source paper’s flat- Λ CDM parameters are used only to identify the distance convention of its published physical quantities. The CPTG quantities reported here are deterministic evaluations of the equations, source assumptions, and comparison coordinates stated in the manuscript. Supporting CPTG code and research references are maintained at <https://github.com/CLG2025/CPTG>. Numerical implementations may retain additional guard digits for reproducibility; manuscript values are rounded to the precision appropriate to interpretation.

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